

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
2	(a)		Baryons: Combination of 3 quarks (1) Accept combination of 3 antiquarks. Don't accept baryons made up of 3 quarks and antiquarks Mesons: Combination of 1 quark and 1 antiquark (1) Don't accept 2 quarks	2			2		
	(b)	(i)	$u\bar{u}$ or $d\bar{d}$ or $\frac{u\bar{u}-d\bar{d}}{\sqrt{2}}$	1			1		
		(ii)	Baryon number LHS: 2, RHS: 1 Particle x must have baryon number: 1 (1) i.e. $1 + 1 = 1 + B_x + 0$ Charge number LHS: +2, RHS +1 Particle must have charge +1 (1) i.e. $1 + 1 = 1 + Q_x + 0$ Particle x is a proton (1) Accept Δ^+ Alternative response to baryon analysis: (1st mark) LHS: uud + uud and RHS: uud + x + $u\bar{u}$ (or equivalent) (1)	1 1	 1		3		
		(iii)	Lepton number is zero on both sides e.g. $0 + 0 = 0 + 0 + 0$ Accept there are no leptons Don't accept there is no change in lepton number	1			1		
	(c)		Electromagnetic (1) γ involvement or photon involvement (1) and 1 reason from: (1) • Lifetime too long for strong or too short for weak (or accept lifetime corresponds to em force) or reference to 8×10^{-17} s or intermediate lifetime. Don't accept quick time or short time • No neutrino involvement [so probably not weak force] • Total u quark number and total d quark number are conserved in the em interaction • Doesn't only consist of quarks [strong force] • No leptons so not weak force Award no marks if incorrect force identified	1 1	 1		3		
			Question 2 total	8	2	0	10	0	0